HOMEWORK 1

Overall 50pts. Submit both markdown and pdf files

You are assigned to analyse a league from the f_data_sm, the leagues are assigned randomly

Sanasar Ananyan - Liga I Sona Avagyan - Le Championnat Saten Harutyunyan - Premier League Sergey Hovhannisyan - Jupiler League Karen Manukyan - La Liga Elen Sahakyan - Bundesliga Narek Sahakyan - Eredivisie Silva Arakelyan - Serie A

```
library(SportsAnalytics270)
library(dplyr)
data(f_data_sm)
```

```
get_top_bottom_teams <- function(data, country, top = TRUE){
  result <- data.frame()
  for (season in unique(data$SEASON)){
    season_standings <- SportsAnalytics270::final_table(data, country, season)
    number_of_teams <- nrow(season_standings)
    if (top == TRUE){
      teams <- season_standings[1:(number_of_teams %/% 2), ]$TEAM
    }
    else{
      teams <- season_standings[(number_of_teams %/% 2):number_of_teams, ]$TEAM
    }
    season_teams <- data %>%
      filter(SEASON == season, HOMETEAM %in% teams, AWAYTEAM %in% teams)
    result <- rbind(result, season_teams)
  }
  return(result)
}
```

1. Calculate average number of goals for home and away teams per SEASON for that league, use dplyr. (4p)
2. Construct a plot using ggplot to show how this number is changing over time. Comment Note: you need to have SEASON on x-axis. Show average Home goals and Away goals on the same plot. Be sure that your plot has appropriate axis names and title.(4p)
3. Interpret the plot (4p)
4. Think of your own approach on how will you measure home team advantage given the data you have. Calculate Home team advantage for your league and for all other leagues over time (4p).
5. Plot the results of the exercise 4 here using ggplot2 and interpret it (4p).

Predictions

1. With your chosen league, predict the probabilities of the first week of the season. Use distribution approach. (4p)

Probabilities

```
data("nba2009_2021")
nba_2021 <- nba2009_2021 %>%
  filter(SEASON_ID == '2021') %>%
  arrange(GAME_DATE) %>%
  select("SEASON_ID","GAME_DATE","home.TEAM_NAME","away.TEAM_NAME","home.PTS","away.PTS", "home.WL") %>%
  mutate(ID = 1:n())

nba_2021_long <-  nba_2021 %>% reshape2::melt(
                    id.vars = c("ID", "GAME_DATE", "SEASON_ID", "home.TEAM_NAME", "away.TEAM_NAME","home
                    measure.vars = c("home.PTS","away.PTS"), variable.name="H_A") %>%
                    arrange(ID)

nba_2021_long <- nba_2021_long %>%
  mutate(TEAM1 = ifelse(H_A == 'home.PTS', home.TEAM_NAME, away.TEAM_NAME),
         TEAM2 = ifelse(H_A == 'home.PTS', away.TEAM_NAME, home.TEAM_NAME))
```